BRAINWARE UNIVERSITY
SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES
DEPARTMENT OF COMPUTATIONAL SCIENCES
BACHELOR OF COMPUTER APPLICATIONS /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /
BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) – 2024
SEMESTER – IV

Course Code: BCA40110
Course Name: Operating System
Weekly Contact Hours: 3L
Credits: 3
Total Allotted Hours: 45L

Course Objective: The aim of this course is to provide an in-depth study of operating systems, their design principles, and their role in computer systems. It covers the fundamental concepts, algorithms, and mechanisms used in operating systems.

Pre-requisite(s): Basic knowledge in computer.

Course Outcomes (COs): After the completion of the course, students would be able to:

CO1: Gain a solid understanding of the fundamental concepts and principles underlying operating systems, including process management, memory management, file systems, and scheduling algorithms.
CO2: Apply knowledge of operating system concepts to analyze and solve real-world problems related to process synchronization, memory allocation, resource management, and system performance optimization.
CO3: Develop practical skills by implementing or simulating operating system components such as process schedulers, memory management algorithms, file systems, or device drivers.
CO4: Evaluate and compare different operating system designs and architectures.
CO5: Learn techniques for measuring and evaluating the performance of operating systems, identifying performance bottlenecks, and proposing optimization strategies to improve system efficiency.

Module 1: Basics of Operating System **[4H]**

Overview of operating system functions and goals, history and evolution of operating systems, types of operating systems, user interfaces and system calls.

Module 2: PROCESS MANAGEMENT: **[16H]**
Process concepts, process Attributes, process state, process control block, Swapping ,process Scheduler, process scheduling, process scheduling algorithms.
CONCURRENCY AND SYNCHRONIZATION: Process synchronization, critical section problem, Peterson's solution, and condition variables classical synchronization problems-producer-consumer, readers and writers problem, dining philosophers problem, semaphores.
DEADLOCKS: System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker's algorithm.

Module 3: Memory Management **[12H]**
Memory hierarchy and organization, memory allocation strategies - contiguous, non-contiguous, paging , structure of the page table and segmentation techniques, virtual memory concepts and address translation, demand paging memory management unit (MMU) and page replacement algorithms.

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Module 4: File Systems and Input/output Systems **[8H]**

File system organization and file attributes, file operations, file allocation methods, directory structure and file system implementation, access control and file permissions, I/O devices, controllers, and interfaces, I/O operations and buffering, I/O scheduling algorithms and device drivers, interrupt handling and I/O software layers, disk management and file system I/O.

Module 5: Distributed System **[5H]**

Distributed system concepts and architectures, distributed file systems and distributed process management.

Course Outcomes (COs) and Modules Mapping:

Sl. No	Course Outcome (COs)	Mapped Module(s)	Learning levels (as per Bloom's Taxonomy from K1 to K6)
1.	CO1	M1	K1, K2
2.	CO2	M2	K3, K4,
3.	CO3	M2, M3	K5, K6
4.	CO4	M5	K4, K5
5.	CO5	M2, M4	K2, K4, K5

Text Books:

1. Operating System Concepts, Abraham Silberschatz, Peter B. Galvin, and Greg Gagne, Wiley Publisher.

Reference Books:

1. Operating Systems: Internals and Design Principles, William Stallings, Pearson Education India, 7th Edition, 2013.
2. Modern Operating Systems, Andrew S. Tanenbaum and Herbert Bos. Pearson.